

JFTSE — Remaining Special Items

Fantasy Tennis client reverse engineering, Java server implementation, red/green tests, and native-client validation

Server repository: sstokic-tgm/JFTSE

Base branch/commit: origin/development / 65c3665170edc2d912a3187244de89251a809712

Work branch: reverse-engineering/remaining-special-items

Faulty-change revert: af927924ea3fa2a4cf1259d35f1e153185018d7e

Implementation: 7079844c6d648aad89686dad5852540ffbb5e31a

Client archive SHA-256:

c19ca21b8e2ab091953b2f631e48853b6477400f4d7000682ac7440f9994f12e

FantaTennis.exe SHA-256:

5477f0827acae66976403aecd2e9ebffeb4fa28da1fedae5f9541ec25e336c31

Runtime: Wine 8.0, Win32 prefix, Xvfb :108 at 1280×900×24

Validation date: 2026-08-12 UTC

Executive summary

This branch implements the currently understood functional subset of Fantasy Tennis special items from first principles. The client catalog, JFTSE packet schemas, existing entity/service boundaries, and native runtime were treated as separate evidence classes rather than interchangeable proof.

The implementation adds match-stat effects for necklaces and earrings, per-successful-match consumption, special-slot validation, proposal-card consumption, promotional ring aliases, exact duplicate-stack handling, and a six-calendar-month nickname cooldown. Match settlement is idempotent per game, and proposal-card mutation is transactional with pessimistic row locks.

Proven by executed tests: item-value mapping, mode filtering, packet field serialization, equipment normalization, one-use consumption, exact-row ring behavior, proposal validation/transaction behavior, nickname cooldown, and single settlement under concurrency. The final full reactor and package gates passed.

Observed with the unmodified native client: the exact release JARs reached User Login, authenticated the disposable fixture, displayed `SpecialLab` and the channel list, and entered the Main lobby. A pre-final packaged run rendered all 12 DB-seeded special-item stacks in Magic Pocket.

Not claimed: native proof of special-slot persistence, proposal UI, match-item consumption, match settlement, Battlemon, Personal Board, or items 5/7/8/18/19/20/22. The character and inventory were DB seeded. A visible necklace equip preview did not persist and is explicitly negative evidence.

Evidence classification

Class	Finding
Observed native runtime	Release client reached login → player/channel selection → Main lobby; earlier package rendered 12 seeded stacks.
Static client reverse engineering	Decrypted <code>Item_Special.set</code> defines values, aliases, mode text, one-use-per-game text, proposal cards, and six-month cooldown copy.
JFTSE source baseline	<code>.packet</code> files establish <code>0x1B70</code> , <code>0x251D</code> , and <code>0x251E</code> ; entities/services define persistence ownership.
Compatibility interpretation	Promotional copies behave like canonical items; WIS catalog effects map to the server's WIL stat; successful settlement is the consumption boundary.
Branch implementation	Java behavior and tests in commit <code>7079844</code> .

Scope, provenance, and branch history

Development base and non-destructive removal

The branch was created from `origin/development` at `65c3665170edc2d912a3187244de89251a809712`. The faulty commit requested for removal, `fcdeb89a94bb17337f8b8f499b79c5892370004e`, was already in published development history. It was removed non-destructively by revert commit `af927924ea3fa2a4cf1259d35f1e153185018d7e`; history was not rewritten.

```
7079844 Implement remaining special item behavior
af92792 Revert "fix types in pet packets; ..."
65c3665 origin/development base
```

The faulty commit remains an ancestor as immutable history, but its patch is counteracted by `af92792`. Nothing was pushed during this work.

Client provenance

The client came from <https://www.jftse.com/client/FantaTennis.7z>. The archive and executable matched the known JFTSE hashes on the title page. `FantaTennis.exe` was unmodified. Runtime copies used a proven Win32 Wine prefix; the immutable client copy was not altered.

Implemented item matrix

Indices	Implemented behavior	Evidence level
4	Next nickname change is six calendar months after the last change.	Static client + unit test
23-25	Proposal-only category/index validation; exact owned stack decremented/deleted atomically; <code>Pocket.belongings</code> updated on final unit.	Schema/source + handler/service tests
27-29	+50/+100/+200 HP; active for Battle/Guardian-family modes, excluded from generic/basic packets.	Static client + serialization/domain tests
30-37	STR/STA/DEX/WIL +3/+5; serialized in earring stat fields; consumed after successful supported settlement.	Static client + tests
39-41	Promotional aliases of canonical EXP/Gold/Wiseman rings; exact equipped <code>PlayerPocket</code> row drives reward and consumption.	Static client + 9 exact-row tests
42-46	Promotional +50 HP / +3 STR/STA/DEX/WIL aliases.	Static client + parameterized tests

Items with an existing implementation outside this slice were not rewritten. Unknown or unresolved catalog entries were not guessed.

First-principles model and invariants

```
DB S0
+ authenticated native action / decoded C2S
+ handler authorization
+ transactional service mutation
→ S2C response/inventory update
→ DB S1
→ client-visible state
```

The implementation enforces these invariants:

1. Special equipment is exactly four slots.
2. A nonzero slot must identify an owned `SPECIAL` `PlayerPocket` row.
3. The same pocket row cannot occupy multiple slots; malformed IDs normalize to zero in the authoritative response.
4. An equipped consumable is decremented once per successful supported match settlement, even if duplicate stale slots reference it.
5. Final-unit removal clears every slot referencing that row and decrements `Pocket.belongings` once.
6. A game can begin settlement once (`AtomicBoolean.compareAndSet`).
7. A proposal record and proposal-card decrement/delete share one transaction.
8. Proposal mutation rechecks category, index, count, and ownership after a pessimistic lock; rejection does not mutate inventory.
9. Ring reward lookup and ring consumption use the exact equipped pocket row, not the first stack with a matching catalog index.
10. Battle-only necklace HP cannot leak into generic/basic status packets.

Documentation and static client basis

The following project documentation was reviewed on 2026-08-12:

- [All wiki pages](#)
- [JFTSE Roadmap](#)
- [Packet Structure](#)
- [Database Schema & Cheatsheet](#)
- [Items](#)
- [Stats](#)
- [Packet Schema format](#)

The wiki informed terminology and navigation but was not treated as runtime proof. The detailed static findings are preserved in `static-client-findings.txt`.

Important static findings

The decrypted client resource identifies proposal-card indices 23-25; necklace values 27-29; earring values 30-37; aliases 39-46; per-game consumption text; and a six-month nickname interval. Japanese necklace text is the strongest static mode qualifier and explicitly names Battle, Battlemon Battle, and Guardian Battle.

Conflicting localizations exist. For example, some necklace English/Thai text uses imprecise material names, and Chinese nickname text says three months while English, German, French, Italian, Japanese, Taiwanese, and Thai text say six. This branch chooses six calendar months because it is the dominant client resource contract and the requested compatibility behavior; that is not a claim about an unavailable original retail backend.

Reverse-engineered protocol

JFTSE packets use the documented eight-byte header followed by packet-specific payload. No new packet IDs were invented.

ID	Direction	Listener	Payload	Use in this branch	Native status
0x1B70	C2S	game/chat	four repeated <code>int32</code> special pocket-row IDs	Validate, normalize, persist, echo authoritative slots	Not observed in the attempted native preview
0x251D	C2S	game	receiver string, pocket-row ID, catalog index, message	Proposal validation and atomic card use	Not observed natively
0x251E	S2C	game	signed status byte	Existing proposal success/error response	Not observed natively

Exact source declarations are archived in [packet-contracts.txt](#).

Listener ownership

Listener	Port	Role in this validation
auth	5894	Login, character list, nickname change
game	5895	Lobby, inventory, proposal, room/match settlement
relay	5896	Started and listening; no special-item claim
chat	5897	Started and listening; mirrors equipment/stat serialization paths
AC	3724	Packaged but not used or claimed in this native run

Java implementation

Match-stat effects and packet fields

`SpecialItemEffects` is the shared mapping for indices 27–37 and 42–46. It writes isolated `specialAddHp`, `specialStrength`, `specialStamina`, `specialDexterity`, and `specialWillpower` components into `EquippedItemStats`. Keeping the components separate prevents a necklace from silently becoming permanent base HP.

Game and chat `FTPlayer` load exact equipped special rows. Existing status, cloth, room, and result packets now serialize earring fields. Generic/basic packets write zero for special HP; battle result/stat paths use active necklace HP. Battle and Guardian state construction adds the isolated special stats to base/clothing/enchantment totals.

Consumption and settlement

Basic, Battle, and Guardian settlement handlers call `MatchSpecialItemUse` only after reaching successful player reward settlement. `MatchplayGame` guards settlement with one compare-and-set operation. The service consumes each distinct equipped active row once, sends updated count/removal packets, and clears exhausted slots.

No native match was completed in this thread. This behavior is proven at the service/settlement test boundary, not through client gameplay.

Promotional rings and duplicate stacks

Indices 39/40/41 are routed to the existing EXP/Gold/Wiseman ring classes. Both reward detection and consumption now iterate/receive exact equipped `PlayerPocket.id` values. This fixes the case where two stacks share one item index and a repository's arbitrary first row is not the equipped row.

Proposal cards

The handler performs cheap request checks and relationship checks; the service then locks the requested inventory row and revalidates category, index, count, and pocket ownership in the mutation boundary. A final unit deletes the row, locks/updates the pocket count, and creates the proposal in the same transaction. Failures map to existing `MSG_SendProposal` status values.

Nickname cooldown

The former fixed 30-day calculation is replaced with `Calendar.MONTH + 6` in UTC. A January 31 boundary test expects July 31, preserving calendar-month semantics rather than approximating six months as a fixed number of days.

Persistence and fixtures

No production schema migration or catalog seed was needed. New values are derived at runtime from existing `PlayerPocket` and `SpecialSlotEquipment` rows. Repository lock methods were added for proposal mutation.

The native account, character, pocket, equipment row, and 12 items were directly seeded into the disposable lab database. They are not client-created evidence. Full fixture and cleanup state is in [native-db-state.txt](#).

Fixture group	Rows
Proposal cards	23×2, 24×1, 25×1
Necklaces/earrings	27, 29, 30, 32, 34, 36; each durable count 2
Promotional rings	39, 40, 41; each durable count 2
Pocket	12/150 belongings
Authoritative special slots	0, 0, 0, 0

Red/green testing

The compact evidence transcript is [red-green-excerpts.txt](#).

Meaningful red contracts

The first focused red test failed because `SpecialItemEffects` did not exist. A later correctness red run caught two substantive regressions:

```
Promotional aliases: expected true but was false (3 cases)
Generic status HP: expected 220 but was 420
BUILD FAILURE
```

The second value proves the test discriminated between ordinary equipment HP (20) and an incorrectly leaked battle-only necklace (+200).

Green matrix

Test class	Final count	Contract
<code>SpecialItemEffectsTest</code>	28	Values, aliases, aggregation, mode activation
<code>SpecialSlotEquipmentServiceImplTest</code>	5	Four slots, ownership/category, duplicate rows, consumption/removal
<code>SpecialItemPacketSerializationTest</code>	1	Exact packet length/offsets and no generic necklace HP
<code>PromotionalRingAliasTest</code>	9	Alias type, reward detection, exact-row consumption
<code>ProposalServiceImplTest</code>	6	Transactional decrement/delete, ownership, locks
<code>SendProposalRequestHandlerTest</code>	8	Statuses, no-mutation rejection, success notifications
<code>MatchplayGameSettlementTest</code>	1	Exactly one concurrent settlement
<code>PlayerNameChangeHandlerTest</code>	1	Six-calendar-month boundary

Post-commit focused rerun: **58 tests, 0 failures, 0 errors.**

Release gates

```
mvn test
game-server: 65 tests, 0 failures, 0 errors
auth-server: 1 test, 0 failures, 0 errors
all 11 reactor modules: SUCCESS
BUILD SUCCESS
```

```
mvn package -DskipTests
all 11 reactor modules: SUCCESS
BUILD SUCCESS
```

Release JAR hashes and runtime details are in [release-runtime-summary.txt](#).

Native runtime walkthrough

The final release smoke run used the exact packaged JAR hashes, the unmodified client, Wine 8.0, a Win32 prefix, Xvfb :108 , Openbox, loopback routing, and supervised Amp services. Packet capture began before login and stopped after the client.

Step 1 — exact release JARs reach User Login



Figure 1 — Unmodified client at User Login after release services started

The ID field, password field, and LOG IN button are visible. No credentials are shown. The initial green unused region is a Wine/window geometry artifact, not server behavior.

Step 2 — authenticate and list the seeded character



Figure 2 — SpecialLab level 1 with Chat and Free channel choices

The native client authenticated against auth port 5894 and rendered the DB-seeded `SpecialLab` character. `Free Channel #1 (Good)` was selected. This proves final-JAR auth/client compatibility; it does not prove native character creation.

Step 3 — connect to the game listener

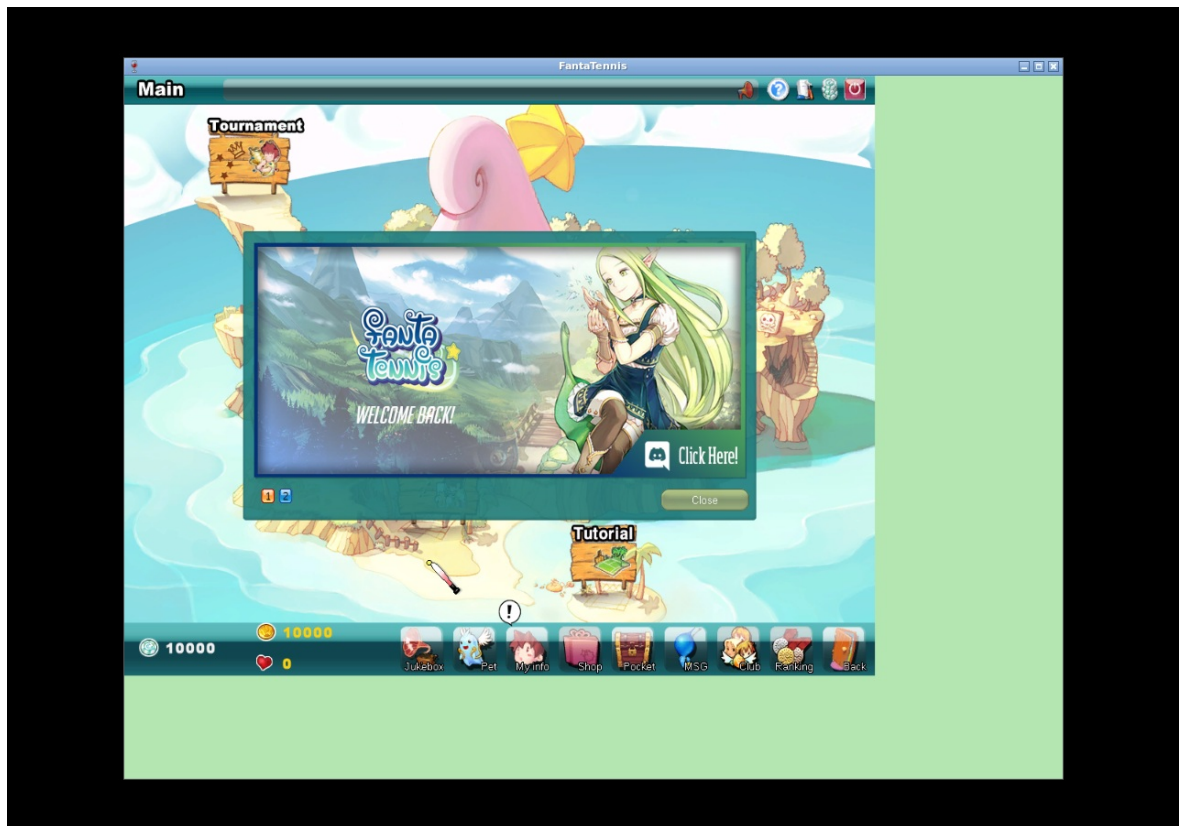


Figure 3 — Main lobby reached through the exact release game JAR

The client transitioned to game port 5895 and rendered the Main lobby. The raw release capture contains one sustained game conversation (344 frames, 27 kB, 756.6 seconds). It is not committed because login traffic can contain plaintext disposable credentials.

Step 4 — render the DB-seeded special inventory



Figure 4 — Pre-final packaged run: Magic Pocket Item/Special shows 12 stacks

This screenshot predates the final exact-row ring backend correction. It shows the same unmodified client and disposable DB fixture rendering 12/150 special stacks: proposal cards, necklaces, earrings, and promotional rings. The final smoke re-proved login/player-list/lobby compatibility after the tiny exact-row change, but did not reopen the pocket successfully. Therefore this figure is inventory-rendering evidence, not final-JAR functional evidence for effects.

Step 5 — negative special-slot experiment



Figure 5 — Client-local necklace preview; explicitly not persistence proof

The UI shows the gold necklace in the first visual slot and HP 400. However, the post-action database remained `0,0,0,0`, and the owning log had no received `0x1B70`. This is a client-local preview/failed emission, not a positive equip result. It is retained to prevent screenshot-only overclaiming.

Native evidence result table

Checkpoint	UI	Packet/listener	DB	Verdict
Release login	User Login	auth handshake on 5894	Fixture available/offline before login	Proven
Release auth	SpecialLab/channel list	successful auth conversation	Disposable fixture	Proven
Release game transition	Main lobby	sustained game:5895 conversation	player active, later cleanly offline	Proven
Special inventory	12 visible stacks	Earlier packaged run	12 exact seeded rows	Proven only as rendering of seeded data
Special equip preview	Necklace shown, HP 400	no received <code>0x1B70</code>	slots stayed all zero	Negative / not proven
Proposal	Not exercised	no native <code>0x251D</code>	no native mutation	Not proven natively
Match consumption	Not exercised	no completed native match	no native mutation	Not proven natively

Packet, DB, and artifact evidence

Release capture boundary

The raw final capture SHA-256 is `f3baa146e6be08785cfad4f72a6e2be11acbfbcd2333fe1dd9cc42acd9782b6e`. It is omitted from Git because protocol payloads can expose credentials. The committed runtime summary records endpoint roles, sizes, conversation totals, JAR hashes, and the capture hash without publishing payloads.

Evidence inventory

File	Source	Purpose	Claim supported
01-release-login-screen.png	Native final run	Login UI	Exact client and release stack reach login
02-release-player-list.png	Native final run	Character/channels	Successful auth and fixture rendering
03-release-game-channel.png	Native final run	Main lobby	Auth-to-game transition
04-special-items-pocket.png	Native pre-final run	Seeded special inventory	Client renders all 12 fixture stacks
05-special-slot-preview-negative.png	Native pre-final run	Failed equip experiment	UI preview must not be treated as persistence
native-db-state.txt	MariaDB snapshots	Fixture and post-preview state	Seed disclosure and all-zero slots
packet-contracts.txt	Repository schemas	Exact payload declarations	0x1B70 , 0x251D , 0x251E contracts
static-client-findings.txt	Decrypted client resource	Catalog semantics	Values, aliases, modes, consumption text
red-green-excerpts.txt	Maven runs	Red/green proof	Missing/wrong behavior then release green
release-runtime-summary.txt	Final native run	Runtime and capture metadata	Exact JAR/client provenance
artifact-sha256.txt	Curated artifacts	Integrity	Screenshot hashes

Compatibility interpretations and limitations

1. **Promotional aliases.** Identical icons/descriptions make 39/40/41 and 42–46 compatibility aliases of canonical rings/stat items. Retail server implementation history remains unknown.
2. **WIS → WIL.** The client resource says Wisdom/WIS while this server and UI use Willpower/WIL. The branch maps the catalog field to WIL.
3. **Consumption boundary.** “Per game” is implemented at successful supported result settlement, not room start or disconnect. This prevents charging a use for an uncompleted room.
4. **Battlemon.** Static text names Battlemon and the effect mapper recognizes that mode, but no native Battlemon lifecycle was validated here. No complete Battlemon support claim is made.
5. **Nickname date arithmetic.** Six calendar months follows dominant client text; it is not a fixed 180-day approximation.

Explicit non-claims

- No native positive or negative proposal walkthrough.
- No native successful special-slot packet or reconnect persistence proof.
- No native completed Basic/Battle/Guardian match consumption proof.
- No multiplayer, relay synchronization, Battlemon, or Personal Board proof.
- No functionality claim for unresolved indices 5, 7, 8, 18, 19, 20, or 22.
- No claim that the original retail server used these exact transactions or error mappings.
- No claim that DB-seeded character/items were created by the client.

Failures and corrected procedure

1. A clean final native launch first displayed only the JFTSE background; after waiting for Wine initialization, User Login appeared.
2. GUI input dropped/duplicated initial characters in two login attempts. The server correctly returned invalid-user status. Separating focus, clear, text, tab, and enter helper calls produced the exact username and a successful third attempt.
3. A final pocket navigation attempt opened Shop/My Info rather than Magic Pocket because custom-cursor and host-cursor coordinates diverged. It was not spliced into positive evidence; the earlier genuine pocket screenshot remains labeled pre-final.
4. A necklace looked equipped in the UI but never produced authoritative packet/DB state. The report classifies it as a failed experiment rather than treating appearance as proof.

Reproduction guide

1. Check out `reverse-engineering/remaining-special-items` and verify commits `af92792` and `7079844`.
2. Use JDK 21 and run `mvn test`, then `mvn package -DskipTests`.
3. Verify release JAR hashes against [release-runtime-summary.txt](#).
4. Start MariaDB and RabbitMQ, then supervised auth/game/relay/chat services.
5. Verify listeners 5894/5895/5896/5897 and DB-advertised endpoint rows.
6. Seed a disposable account/character/pocket/items; record that this is a DB fixture. Reset login/online state before each run.
7. Point CRLF `ServerInfo.ini` to `127.0.0.1:5894`.
8. Start packet capture before interaction. Use a distinct Win32 Wine prefix, Xvfb display, Openbox service, and runtime copy per native client.
9. Drive `FantaTennis.exe` from its runtime working directory; verify each UI barrier and pair screenshots with packet and DB checks.
10. Stop client and capture cleanly; inspect PCAP for credentials before any publication. Recheck account offline state.
11. Build this report with the committed stylesheet and visually inspect every rendered page.

Final conclusion

The branch now contains a coherent, tested server implementation for the understood remaining special-item slice. All final tests and package gates are green, exact release JARs interoperate with the unmodified client through the Main lobby, and the report preserves both positive visual evidence and the critical failed equip boundary.

The strongest functionality claims are test-backed server contracts. Native runtime evidence is deliberately narrower: connectivity, character/channel rendering, lobby transition, and seeded inventory rendering. Proposal, successful equip persistence, consumption, and match/Battlemon flows remain future native experiments rather than claims in this work.